

AI, AGI, Math and Truth

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June 9, 2026

Recent years we have witnessed a flourishing of AI technologies. Optimists believe that we are experiencing an information revolution led by AI; similar to the agricultural revolution and industrial revolution, we are at a pivotal moment in history [2]. Pessimists believe that, just like the spread of printing in Europe gave rise to witcher hunting and the spread of telegraph and television gave rise to the Nazi regime [3], the advancement of information technology does not necessary promise a better future, until we have learned how to properly use them.

In this article, I would like to share some of my opinions on AI. Personally, I have been working on theoretical AI research for a few years. This article will be written in a story-telling rather than a technical report style, and I will answer the questions mainly from principal and philosophical aspects.

1. What is AI? And how does it work?

Suppose we have a set of images, some of them are cats and some of them are dogs. How can we teach a machine to distinguish which image is of cat and which is of dog? We know that digital images are formed by a collection of pixels, and we may assume that there are two distributions P and Q over these pixels, where P represents cats and Q represents dogs. Given an image x , if $P(x) > Q(x)$, we say x is of cat; otherwise we say x is of dog.

The *underlying* distributions P and Q can be irregular and we do not know how they look like. However, if we can make the machine to calculate an *artificial* function f that *approximates* the density function of P or Q , then we can use f , instead of P and Q , to make a prediction for x . The critical question is that, how to construct such an f ?

For a long time scientists have no idea until the breakthrough at the ImageNet Challenge in 2012. Alex Krizhevsky, a student of Geoffrey Hinton, constructed this f using a multi-layer artificial neural network called Alexnet, and calculated it with the assistance of a NVIDIA graphic card. Surprisingly, this approach won the challenge, as well as significantly outperforming the runner-up.

The success of Alexnet in image data quickly drew attention among researchers. Broadly speaking, almost all human related data – including images, languages, audios, videos and gaming decisions – can be viewed as statistical distributions. People began to wonder whether artificial neural network is a universal approximator for all such distributions; if so, its success in images can be transferred to languages or other types of data. They got it right this time; as a result, we witness the flourishing of AI technologies in various fields.

In summary, current AI technology is an application of statistics. Its goal is to construct some artificial functions to approximate the underlying distributions of human data. Artificial neural networks have been shown to be the most effective way to construct such functions,

and they are the foundation of most modern AI technologies.

2. What is AGI? And how does it differ from AI?

Artificial General Intelligence (AGI) can be considered the ultimate form of AI that simulates the full cognitive ability of humans. Human cognitive ability includes not only *intelligence*, but also *consciousness*. Current AI has intelligence, but it lacks consciousness [4, 3].

Intelligence is the ability to make complex decisions to attain goals. It may exist without consciousness. For example, bacteria and plants apparently lack consciousness, yet they display intelligence for obtaining food, reproducing, and cooperating with other organisms [3]. Intelligence can exist even when it detaches from the organism, as AI is an inorganic computer program that accomodates intelligence.

What about consciousness? Philosophically, consciousness is something that exists but we may not be able to use language to precisely describe what it is. Consciousness has many manifestations, such as “the ability of experiencing subjective feeling such as pain and pleasure” [3], “having free will” [4], “being able to understand the world” [4]. I will explain why current AI does not have these features.

Let us first talk about understanding. Suppose humans designed a system of decision rules, and an AI can learn all these rules so that it can make desicions at the same level of a human. How can we tell whether AI understands these rules, or it simply memorizes these rules? Understanding rules means that an entity can tell *why these rules work or not work* – this knowledge lies outside of the system of rules and is not achievable by simply following the rules [4]. This is crucial in creative thinking, since understanding why these rules work give you something the rules don’t give you. Current AI can discover what we have omitted within the system of rules that human predesigned for it, but none of them can tell some truth outside of the system.

Then we talk about free will. We say something has free will if its behavior is unpredictable even after we gain the full knowledge of it [4]. Roger Penrose believed that the free will of creatures origins from quantum mechanic, since quantum is the only object in modern physics that is theoretically unpredictable, by Heisenberg’s indeterminacy principle [4]. The behavior of current statistic-based AI is predictable; in fact, scientific approaches – theory and engineering – aim at making things certain and predictable. Thus, when we build an AI under scientific approaches, its unpredictable nature is excluded at the very beginning.

3. What are the obstacles in achieving AGI?

Recall the evolution of life: Nature seems arrives at the construction of consiousness first – which appears common at higher animals; only after that, nature builds language, which humans possess. In this sense, AGI is indeed a reverse engineering: humans are trying to build consciousness starting from language [4].

Is AGI achievable within the current statistical framework of AI? It seems that none of the components of current AI can accomodate consciousness. So, when these components are assembled, why should we believe that consciousness will naturally emerge? I think the belief on “consciousness will emerge from nothing” is pseudoscientific, similar to Homunculus – a medieval belief that a small artificial human can be created in a bottle through alchemy.

Thus, to achieve AGI, current AI should at least integrate some prerequisite technologies that perform consciousness [4]. If Penrose’s quantum consciousness theory is correct, such

technologies must be based on quantum mechanic. Quantum computing could be such a candidate technology, however, how to make it practical at a large scale and how to link it to consciousness are unsolved problems at current stage.

4. Can AI distinguish truth from information?

Let me first explain the difference between information and truth. There are several different definitions on information, such as “a new reality that will possibly happen in the future” [3]; I will instead focus on the mathematical one. Back to 1940s, Claude Shannon established the information theory, which build the connection between information and statistics. The information of events can be characterized by a random variable, and an event x that occurs with probability $p(x)$ has the amount of information $-p(x) \log p(x)$, i.e., a lower probability implies a higher amount of information. This genius insight illuminated the path for processing information via statistics, which is why current AIs are typically statistical models.

Truth, according to the ancient Greek philosopher Parmenides, is *what is unchangeable in this changeable world* [1], such as “the sun rises from east” and “all humans will die”. Information does not necessary represent the truth; we say information is true only when it accurately reflects reality.

Before discussing whether AI can distinguish truth from information, consider how humans distinguish truth. Are things that most people believe to be true necessarily true? Before Darwin’s evolution theory, most people believe that humans are directly created by God rather than being evolved from monkeys, which is not true in a modern view. Are things told by professionals or authorities necessarily true? Even Einstein could make mistakes, as he disbelieved the stochastic nature of quantum and said “God does not play dice”.

In general, humans themselves are not good at recognizing truth [3]. They are pretty much like *blind men appraising an elephant*, each of them may obtain a partial truth, but none of them can see the entire truth. No one can guarantee that they do not make mistake. Experts can be wrong and outliers can be right.

Unlike information, there is neither a scientific metric to quantify the amount of truth, nor a guidance that guarantees reaching truth. But, as philosophers suggested, maintaining a sharp understanding about this world usually gives humans a good eyesight to identify truth. Such a sharp understanding is unlikely possessed by an unconscious AI, which has no motivation to understand this world.

5. Can AI surpass humans in proving mathematical theorems?

Recently, OpenAI’s newest reasoning model disproves a 80-year-old conjecture [5, 7] – the planar unit distance problem, proposed by Paul Erdős in 1946. The model unexpectedly uses tools from a less related field – algebraic number theory, to construct a counterexample that solves this discrete geometry problem. Nearly no mathematicians have tried this before. The Fields medalist Tim Gowers stated that this is a milestone in AI mathematics and the proof is professional enough to be accepted by Annals of Mathematics [7].

Despite the exciting breakthrough, public wonders that: Why does the reasoning ability of AI become so powerful? Does this mean AI understand math better than human? Will Riemann hypothesis be eventually solved by AI? Will mathematicians lose job in the future?

I would say that AI reasoning is not that magical as the public believes. The key technology behind it – the proof assistants such as Lean [6, 8] – is not new. Before the rise of AI, these

tools had been developed to assist mathematicians completing proofs. Take Lean for example: mathematicians first formally write proofs in Lean codes, then use Lean to execute these codes to check their correctness.

What is new in AI-generated proof? Remember that AI can write codes. Humans first train AI to write Lean codes, then use Lean to execute AI-generated codes. Since AI replaces human to interact with Lean, Lean can generate proofs much faster – this is why we see a rapid progress of AI-generated proofs. However, this proof-discovering approach is essentially a brute-force search algorithm. Neither AI nor Lean necessarily understands math concepts: they just manipulate symbols according to the rules that humans designed.

Why is Lean important? Without Lean, AI by itself cannot guarantee the correctness of a proof. The statistical nature of AI determines that it can recognize “what is related with high probability”, but cannot recognize “what is related absolutely” – absolute is deterministic, not probabilistic. A mathematical proof is a set of logical chains, and logics are absolute, which cannot be recognized by AI.

Does this proof-discovering approach have limitations? As long as this approach is algorithmic, it cannot discover all proofs. This conclusion was carried out last century. Mathematicians found that some math conjectures can be solved by algorithms – the most famous one is the *four color theorem*, solved by Appel and Haken using brute-force searching in 1976. They also found some math problems that no algorithm can solve, such as determining whether a Euclidean plane can be covered by a finite set of polyominoes [4].

Will Riemann hypothesis be eventually solved by AI? We don’t know. We don’t even know whether it is a reachable truth – by Gödel’s incompleteness theorem, if a math system is complete and consistent, then there is some truth that requires infinite steps of deductions to reach, but neither human nor machine can produce infinite steps of deductions.

References

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